
DeepFLAIR*: Evaluating U-Net Architectures for EPI-to-FLAIR* Synthesis

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Abstract

FLAIR* is a synthetic MRI contrast generated by combining two separate acquisitions, and can be generated from a single scan using deep learning. Current state-of-the-art image synthesis in medical imaging uses a standard U-Net. We compare three transformer-based U-Net architectures (Attention U-Net, Swin-UNETR, and UNETR++ with Voxel-Focused Attention) to a baseline U-Net on a dataset of 89 subjects using a composite loss of MSE, SSIM, and gradient consistency. All models except UNETR++ achieve comparable test SSIM (≈ 0.918 – 0.924), while UNETR++ produces visibly blurred outputs due to overly restricted attention windows. Our findings suggest that in medical image synthesis tasks, dataset size, memory constraints on batch size, and loss function design are dominant performance factors, consistent with conclusions drawn in the segmentation literature. Future work should prioritize patch sampling strategies and loss weighting over architectural novelty.

1 Introduction

Multiple sclerosis (MS) is a chronic inflammatory disease which presents lesions in the brain. These lesions can be seen in Magnetic Resonance Imaging (MRI), and are particularly visible in T2-FLAIR scans, where they appear as hyperintense (brighter) against white brain matter [1, 2]. The Central Vein Sign (CVS) is a key biomarker which can be used to differentiate MS lesions from common mimics [3, 4]. CVS can be most clearly seen in T2*-weighted magnitude 3D-EPI images, in which veins appear hypointense. By combining the T2*-weighted and T2-FLAIR through image multiplication, we can generate a FLAIR* image which demonstrates hyperintense lesions with dark central veins [5, 6].

A single high quality MRI scan can take 25 minutes [7]. Movement during and between scans can lead to artifacts and misregistration, and long periods in the scanner may not be accessible to all patients. Workflows which require lengthy scans, combine multiple acquisitions, or involve offline post-processing can restrict implementation in clinical settings. Emerging work has offered deep learning solutions through image synthesis tasks such as super-resolution [8] and synthetic contrasts [9, 10].

Producing an accurate FLAIR* from the T2*-weighted magnitude images is particularly difficult as it requires learning to suppress the hyperintense cerebrospinal fluid (CSF) to hypointense while maintaining fine details of the CVS. Regression-based encoder-decoder architectures, such as the U-Net, are well suited to this task. These architectures provide the depth to learn complex features while preserving details through skip connections. In this paper, we compare the performance of four U-Net architectures on the FLAIR* synthesis task.

2 Methods

We used the CAVS-pilot dataset (N=89) with a train/val/test split of 0.7, 0.1, and 0.2 respectively. FLAIR* volumes were generated from T2* 3D-EPI and T2-FLAIR acquisitions through linear image registration, interpolation, and image multiplication [5]. Each volume was individually intensity-normalized to $[0, 1]$, and padded to $(320, 384, 320)$ voxels.

For training, $(64, 64, 64)$ voxel patches were extracted randomly and augmented with random flipping in the axial and sagittal directions. Gaussian blurring was applied at three discrete levels (none, weak, medium) with equal probability. Augmentation was not applied during validation or testing. Patches were passed into each architecture with subject multiplication to balance epoch lengths to 32 steps (Table 1). During inference, sliding window patches of size $(64, 64, 64)$ were taken at stride 32, and blended with Gaussian overlap.

Loss was calculated through a composite objective function combining mean-squared-error (MSE) and structural similarity (1-SSIM) [11]. A gradient consistency term was also included, calculated as the L1 loss of finite differences in each spatial dimension, averaged across axes. Gradient clipping ($\text{norm} \leq 1.0$) was applied. All loss weights were set equally to 1.0. Models were optimized with Adam ($\text{lr} = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$).

Models were trained until plateauing or reaching 300 epochs. The checkpoint producing the lowest validation loss was used for evaluation on MSE and SSIM. All experiments were conducted on an Ubuntu workstation with a 32-core 3.7 GHz CPU, 256 GB memory, and four NVIDIA Quadro RTX 6000 GPUs.

In total, four architectures were tested: the baseline U-Net, Attention U-Net, Swin-UNETR, and UNETR++ with Voxel-Focused Attention. The U-Net and Swin-UNETR were additionally evaluated with a sigmoid final activation.

Table 1: Training configuration per architecture.

Architecture	Batch Size	Subject Mult.	Steps/Epoch	Params
U-Net	16	8	32	5.6M
Attention U-Net	16	8	32	5.7M
Swin-UNETR	2	1	32	15.7M
UNETR++ (VFA)	32	16	32	1.2M

2.1 Baseline U-Net

The baseline U-Net from the original DeepFLAIR* paper [10] was replicated in PyTorch. The encoder has four Double Convolutional layers with BatchNorm and LeakyReLU, and average pooling steps in between. The bottleneck is a full Double Convolution layer, and the decoder matches the encoder dimensions with skip connections allowing features to propagate past the bottleneck. The output goes to a $1 \times 1 \times 1$ convolutional block followed by a ReLU activation. We also evaluated a final Sigmoid activation. BatchNorm was used as larger batch sizes are feasible due to the lower memory footprint. In the original paper, the network was trained for 75 epochs using deterministic grid patching and fine-tuned with Optuna at batch size 8.

2.2 Attention U-Net

The Attention U-Net [12] introduces attention gates between the skip connections of the encoder and decoder. These gates learn which encoder features should be propagated or suppressed. For example, they can allow CVS structure to pass through while attenuating hyperintense CSF. The gates are computed as:

$$W_x(\mathbf{x}) + W_g(\mathbf{g}) \xrightarrow{\text{ReLU}} \psi(1 \times 1 \times 1 \text{ conv}) \xrightarrow{\text{Sigmoid}} \alpha \Rightarrow \mathbf{x} \cdot \alpha \quad (1)$$

where $\mathbf{x}$ is the skip connection, $\mathbf{g}$ is the gating signal from the decoder, and α is the learned spatial attention mask. The Attention U-Net is otherwise identical to the baseline U-Net.

2.3 Swin-UNETR

Pure Vision Transformers (ViTs) compare each voxel to all others, giving quadratic complexity in the number of voxels. They also lack the inherent local spatial inductive bias of CNNs, requiring large datasets to learn spatial relationships from scratch.

The Sliding-window U-Net Transformer (Swin-UNETR) [13, 14] limits self-attention to local windows of $(2, 2, 2)$ tokens, while a shifted-window strategy allows each token to interact with new tokens across layers, building a global receptive field. We used a feature size of 24. Due to memory constraints, batch size was limited to 2. BatchNorm is unstable at small batch sizes, so instance normalization was used throughout [15].

2.4 UNETR++ with Voxel-Focused Attention

UNETR++ [16] addresses ViT quadratic complexity through Efficient Paired Attention (EPA) blocks. Each EPA block splits computation into parallel spatial and channel branches, sharing Q and K projections but with separate V:

- **Spatial branch (VFA):** Standard UNETR++ relies on dimensionality reduction which risks information loss. By using Voxel Focused Attention [17], each voxel computes self-attention only within its local $(3, 3, 3)$ neighborhood, reducing complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(27N)$ while preserving representational power.
- **Channel branch:** Each channel is treated as a token, computing a $C \times C$ attention matrix to capture global feature relationships without spatial cost.

The outputs of both branches are then summed and projected. The encoder uses a 3-stage hierarchy with patch size 4, producing 16^3 , 8^3 , and 4^3 token grids, with a standard U-Net CNN decoder and skip connections. The architecture has approximately 1.2M parameters, enabling training at batch size 32 alongside heavier models.

3 Results

Table 2 reports validation loss (the composite training objective) and test-set metrics for all architectures. Test metrics are computed on the held-out test set (18 subjects) on full volumes normalized to $[0, 1]$; SSIM is reported as actual structural similarity (higher is better), and MSE is mean-squared error (lower is better).

Table 2: Validation loss and test-set performance. Best result per column in **bold**. [†]Still training at submission; final results will be posted at <https://github.com/alexandershen27/deepflairstar>.

Architecture	Val Loss ↓	Test MSE ↓	Test SSIM ↑
U-Net (ReLU)	0.0737	0.000869	0.918
U-Net (Sigmoid)	0.0617	0.000914	0.924
Attention U-Net	0.0750	0.000914	0.914
Swin-UNETR	0.0637	0.000901	0.918
Swin-UNETR (Sigmoid) [†]	0.1204	—	—
UNETR++ (VFA)	0.1014	0.001168	0.888

Training curves are shown in Figure 1. U-Net (Sigmoid) achieved the lowest validation loss (0.0617 at epoch 192), followed by Swin-UNETR (0.0637 at epoch 230), U-Net ReLU (0.0737), and Attention U-Net (0.0750). UNETR++ converged to a higher validation loss (≈ 0.10).

Qualitative results are shown in Figure 2. U-Net, Attention U-Net, and Swin-UNETR architectures produce visually plausible FLAIR* predictions with broadly correct CSF suppression, while maintaining some lesion hyperintensity and CVS appearance. UNETR++ outputs are noticeably blurred. Difference maps reveal residual errors concentrated around CSF boundaries and deep white matter.

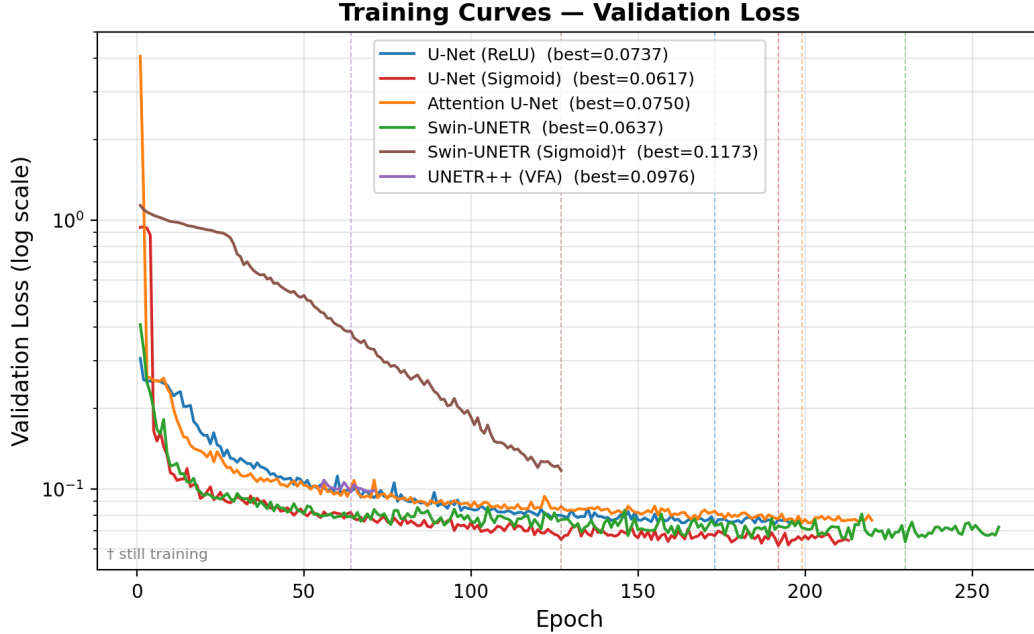


Figure 1: Validation loss over training epochs (log scale). Dashed vertical lines mark each model’s best-checkpoint epoch. †Still training at submission.

4 Discussion

4.1 Architecture Comparison

Across both validation loss and test metrics, the performance gap between architectures was narrow. Swin-UNETR achieved the second-best validation loss (0.0637) and matched U-Net (ReLU) on test SSIM (0.918), despite having nearly $3\times$ the parameters and requiring $8\times$ smaller batch sizes due to memory constraints. Attention U-Net, which adds soft attention gates to the skip connections, did not improve over the plain U-Net on any metric, suggesting that the attention mechanism did not provide meaningful benefit. UNETR++ produced the worst validation loss (≈ 0.10) and visibly blurry outputs, which we attribute to the $(3\times 3\times 3)$ local VFA window being too small to reason about local intensity structure, a limitation that matters more for synthesis than segmentation.

These results echo the conclusions of Isensee et al. [18], who showed that a well-configured U-Net consistently outperforms more complex architectures on medical image segmentation benchmarks. We find that this holds for synthesis as well: the dominant bottlenecks in this task are dataset size, memory constraints that limit effective batch size for transformer-based models, and complex learning tasks such as CSF suppression and CVS preservation, none of which are addressed by architectural novelty alone. The most meaningful performance gain came not from a change in architecture but from switching the final activation from ReLU to Sigmoid, which constrained outputs to $[0, 1]$ and yielded the best validation loss overall (0.0617), as shown in Table 2.

4.2 Loss Functions

The composite loss function for the baseline training used in DeepFLAIR* was manually tuned [10]. We trained models with equal weighting to evaluate the efficacy of each loss component in image-to-image translation.

During training, validation loss was dominated by the SSIM term, while MSE and gradient loss remained relatively low. This likely contributes to difficulty learning CSF hypointensities in the ventricles and brainstem, which require the model to invert the intensity of entire regions. Increasing MSE weight empirically reduced MSE without worsening SSIM or gradient loss.

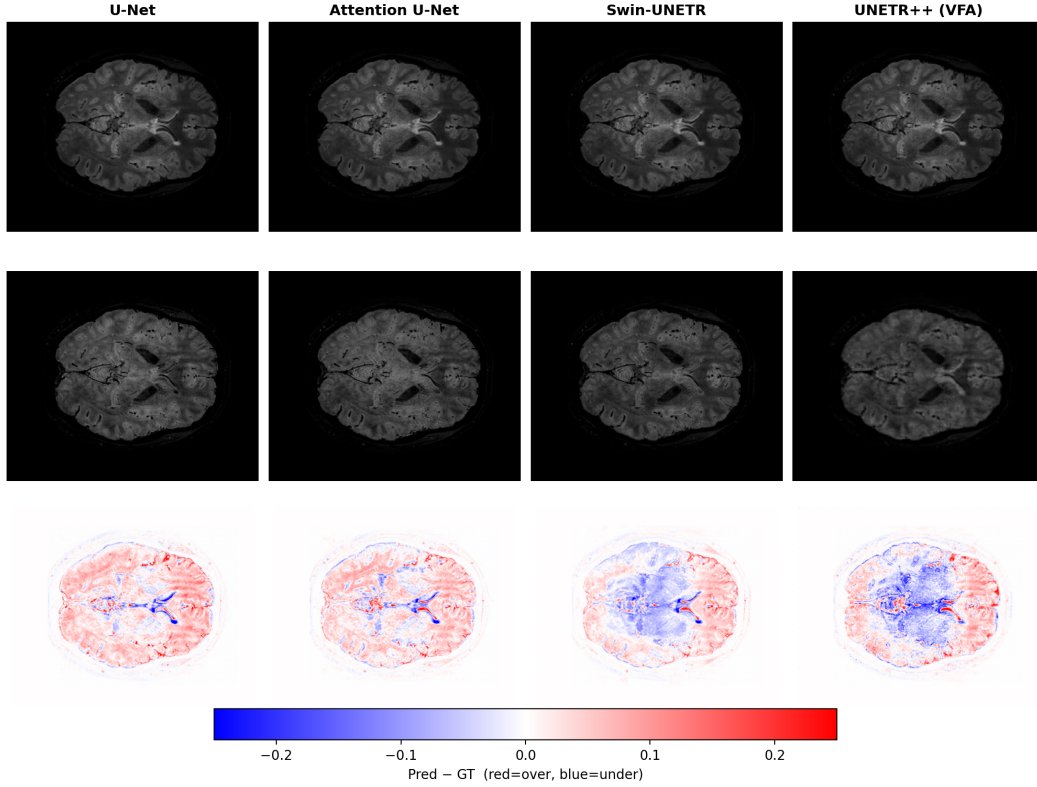


Figure 2: Sagittal slice comparison. *Top*: ground-truth FLAIR*. *Middle*: predicted FLAIR*. *Bottom*: signed difference map (Pred - GT); red indicates over-prediction, blue indicates under-prediction (± 0.25 intensity units).

L1 loss was also tested to encourage learning of black background voxels. An early Swin-UNETR experiment with BatchNorm and sigmoid failed to suppress intensities with L1, while hard sigmoid succeeded (Figure 3). L1 also appeared to push CSF intensities darker than the FLAIR* ground truth.

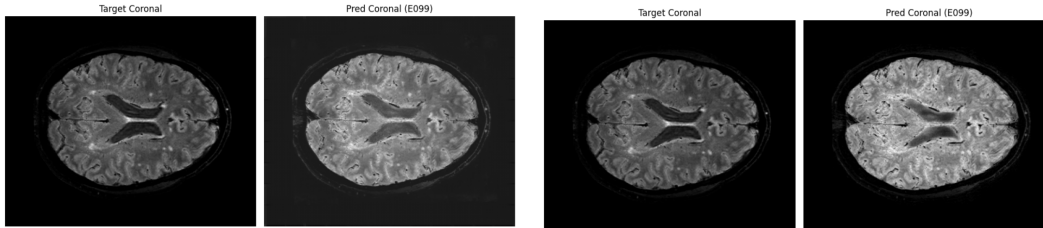


Figure 3: Early Swin-UNETR outputs with BatchNorm and L1 loss. *Left*: sigmoid activation fails to suppress background. *Right*: hard sigmoid successfully suppresses background voxels.

4.3 Final Activation Functions

A ReLU activation is used at the output of the baseline U-Net. This constrains output to positive values but does not cap outputs below 1.0. During validation, predicted voxels occasionally exceeded 1.0. Empirically, this did not appear to affect SSIM loss significantly.

During early training, some architectures produced mostly-black outputs (near-zero activations). Recovery occurred within several epochs, likely due to gradient signal from skip connections. Kaiming initialization may have stabilized early training.

Preliminary Swin-UNETR experiments with instance norm and no final activation successfully trained for 30 epochs with reasonable outputs, suggesting activation functions may not be strictly necessary. With BatchNorm, sigmoid struggled to push background voxels to zero even with L1 loss, while hard sigmoid succeeded.

4.4 Learning Rate and Training Dynamics

The Optuna-tuned Adam parameters from [10] ($\beta_1 = 0.468$, $\beta_2 = 0.94$) appeared too aggressive for learning CSF hypointensities. Models would briefly produce dark CSF patches, spiking the loss, then immediately revert. Standard Adam ($\beta_1 = 0.9$) was used for all reported experiments. No learning rate scheduler was used; manually annealing the learning rate after plateau detection would likely improve final performance.

4.5 Limitations

The original DeepFLAIR* implementation uses deterministic grid patching at stride 32, yielding a fixed number of patches per subject. This was not feasible for Swin-UNETR due to memory constraints and epoch length. Random patching was used instead, which may reduce spatial coverage consistency but could improve generalization.

Gaussian blending was used for sliding window inference. A Hann window would theoretically reduce boundary artifacts but could not be verified within the scope of this work. Optuna hyperparameter search was not applied to all architectures due to compute constraints.

4.6 Future Work

Several architectures could warrant further investigation, including DADC layers (Dual Attention and Depthwise Convolutions) for high-frequency texture preservation, GAN-based training for perceptual sharpness, and diffusion models with appropriate conditioning for stochastic synthesis. A learning rate scheduler (e.g., ReduceLROnPlateau) would likely yield further improvement for all architectures tested.

Beyond architectural changes, improvements to the training pipeline may yield stronger gains. Lesion-biased patch sampling, which weights patch centers toward hyperintense regions, could improve sharpness of MS lesions and CVS structure in synthesis outputs, which are underrepresented in random patches over the full brain volume. Anatomically-masked loss functions, computing MSE or SSIM only within brain tissue or lesion regions, could focus the optimization on clinically relevant areas rather than penalizing background voxels equally. Better tuning of the composite loss weights, particularly increasing the MSE term relative to SSIM, may also help the model learn regional intensity inversions such as CSF suppression.

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